Superconductivity

1. For a certain metal the critical magnetic field is $5X10^3 \ A/m$ at 6K and $2X10^4 A/m$ at 0K. Determine its transition temperature.
   (Ans: $T_c = 6.93K$)

2. Calculate the critical current for a wire of lead having a diameter of 1mm at 4.26K.The critical temperature for lead is 7.18K and $H_c(0) = 6.5X10^4$ A/m. (Ans: $H_c(T) = 4.2118X10^4$ A/m)

3. A Lead superconductor with $T_C = 7.2K$ has critical magnetic field of $6.5X10^3$ A/m at absolute zero. What
   Would be the magnitude of critical magnetic field at 5K temperature?
   (Ans: $H_c(T) = 3.365X10^3$ A/m)

4. The critical magnetic for vanadium field is $10^5 \ A/m$ at 8.58K and $2X10^5 A/m$ at 0K. Determine its critical temperature.
   (Ans: $T_c = 12.133K$)

5. A Lead wire has a critical magnetic field of $6.5X10^3 A/m$ at 0K. The critical temperature is 7.18K. at what temperature the critical field will drop to $4.5X10^3 A/m$?
   (Ans: 3.9827K)